

Random validation optimism is heterogeneous across civil-engineering machine-learning datasets

a What random folds can hide

Random CV

same source families split across train and test



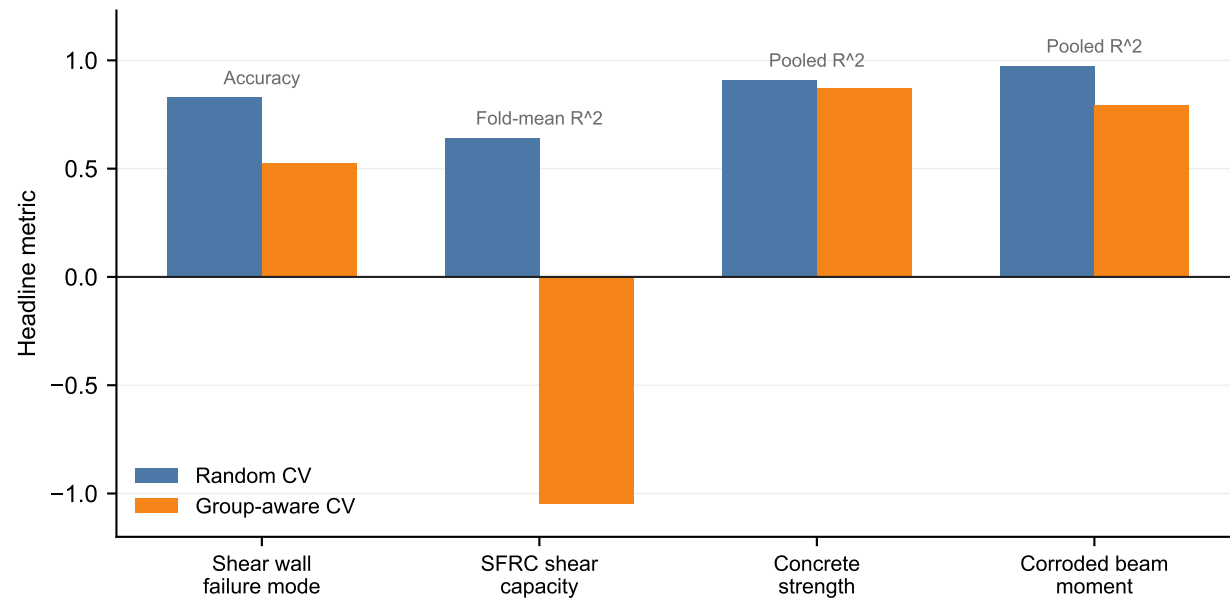
Group-aware CV

entire source, mix or program held out

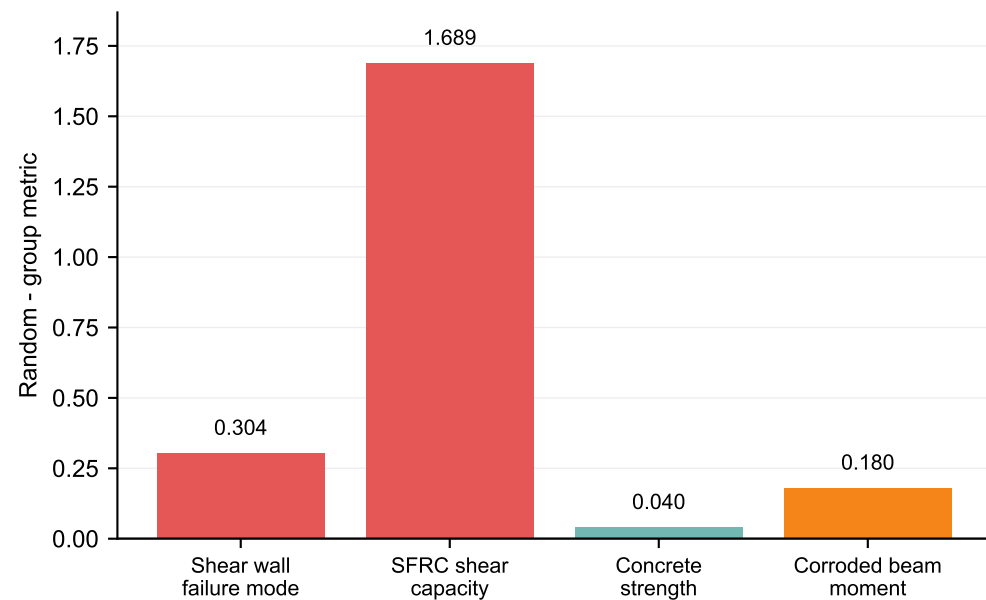


harder
claim
↓

b Random validation gives higher headline scores



c Optimism is a distribution, not a constant



d Dataset topology explains part of the heterogeneity

